KUSHAGRA DWIVEDI

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SUMMARY

Final-year CS student at VIT (8.37 CGPA) who builds things people actually use. Shipped RepoAuditor, an AI audit tool that cut a GitHub repo review from about 3.3 hours to 12 minutes, and a YOLOv8m pothole detector at 91% precision. Interned as a React Native developer at Civilytix, where a pothole reported through my app got fixed by the municipality. Looking for software engineering and AI/ML roles.

TECHNICAL SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript, SQL
Backend: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, JWT auth, async programming
Frontend and mobile: React, React Native, Expo, Vite
AI/ML: YOLOv8, OpenCV, LLM prompt engineering, RAG, vector embeddings, Groq API, Hugging Face, Gradio
Cloud and tools: Docker, Microsoft Azure, Render, n8n, Git, GitHub

EXPERIENCE

Software Engineer Intern, Civilytix Pvt. Ltd. *Jan 2026 - Mar 2026*

- Built the React Native (Expo, TypeScript) frontend for a road-safety app where people report potholes and other hazards with a photo and location, sign in, and see reports on a live map.
- Added a Community Contribution feature. One report submitted through it was confirmed fixed by the municipality, proving the app works from report to repair.

PROJECTS

RepoAuditor: AI-powered GitHub repository audit SaaS

- Cut the time to audit a GitHub repo from **about 3.3 hours to 12 minutes** with a 5-agent n8n workflow that checks architecture, security, performance, and dependencies.
- Added a debate agent that cross-checks the other agents' findings before they reach the user.
- Built the FastAPI backend with JWT login, password reset, user reviews, and Razorpay billing; hosted it on Render with PostgreSQL, and deployed the AI workflow on Microsoft Azure.
- **Stack:** FastAPI, Python, PostgreSQL, SQLAlchemy, n8n, Groq API, GitHub API, Razorpay, Resend, Docker.

Road-Defect Detection System

- Trained YOLOv8m on **3,386 hand-annotated** road images to find potholes and rate their severity from bounding-box size.
- Raised precision from **88.7% to 91%** by preprocessing images with grayscale, CLAHE, Gaussian filtering, and Canny edge detection.
- Built a Streamlit app that runs detection in real time.
- **Stack:** Python, YOLOv8, OpenCV, Streamlit.

Career Conversation Agent

- Cut post-ATS screening time from **26 minutes to 7 minutes** per resume across a 10-resume benchmark.
- Built a chat agent on Groq's Qwen-3-32B using prompt engineering and RAG, served through a Gradio app.
- **Stack:** Python, Groq API, vector embeddings, Hugging Face, Gradio.

EDUCATION

B.Tech, Computer Science – Vellore Institute of Technology *2023 – 2027 (expected) | CGPA 8.37*

Class XII (CBSE) – Delhi Public School, Jankipuram *2023 | 86.2%*

Class X (CBSE) – Delhi Public School, Jankipuram *2021 | 94.1%*

CERTIFICATIONS

IBM Agentic AI Professional Certificate | DeepLearning.AI Machine Learning Specialization | Astronomer Data Engineering Foundations | Google Cloud Generative AI Leader

ACHIEVEMENTS

- Solved 320+ DSA problems on LeetCode.
- Selected for McKinsey's Forward Program.